Transport of Legionella Relevant Aerosols from Showers

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Disclaimer

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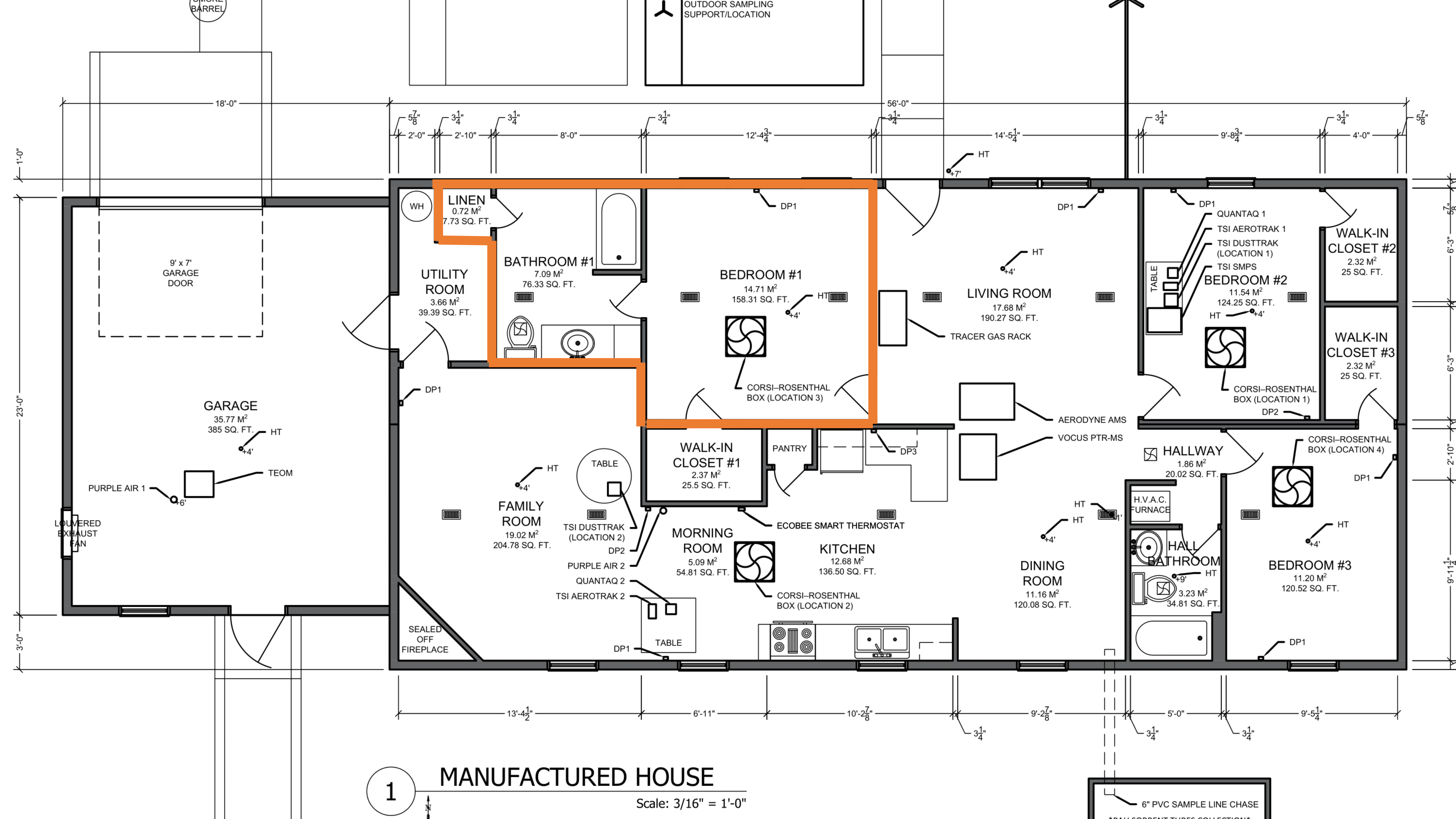
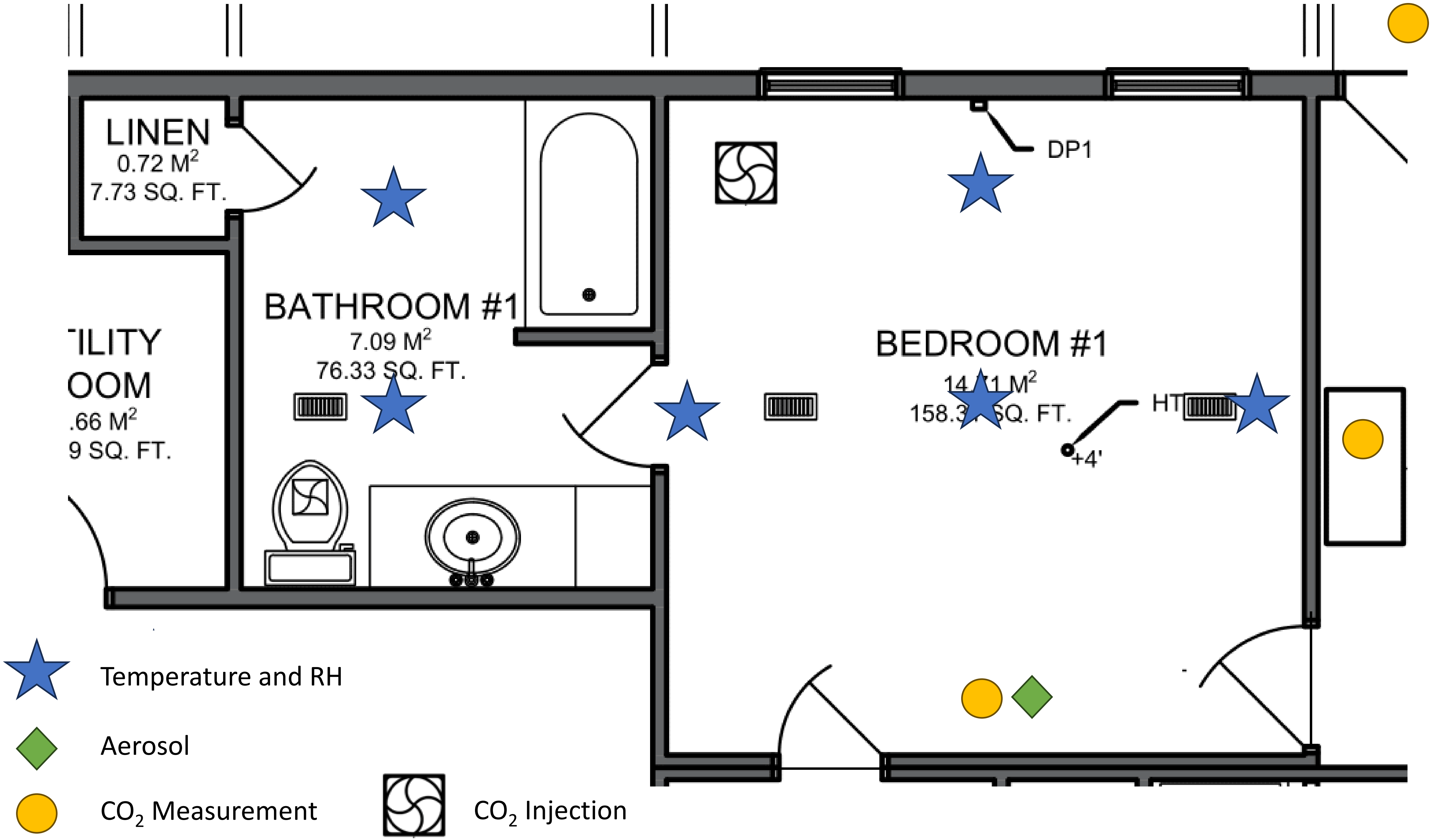
Manufactured House



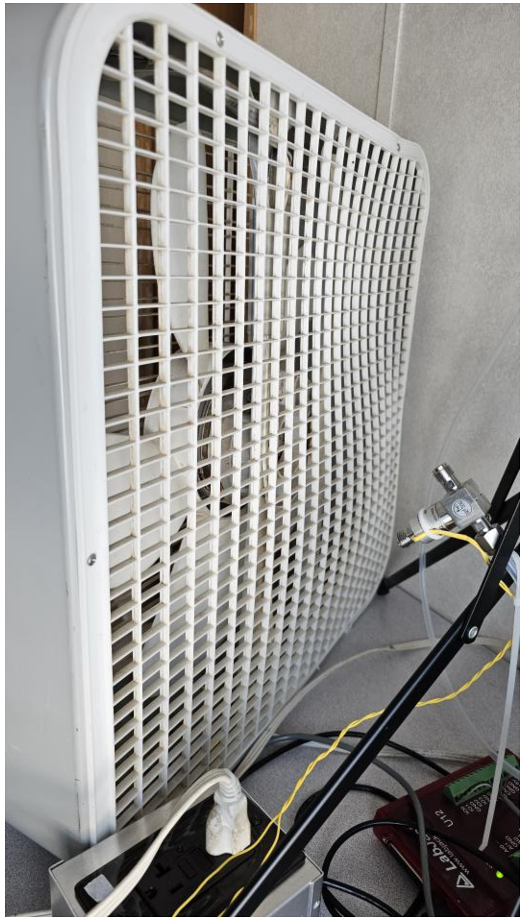
1 MANUFACTURED HOUSE

Scale: 3/16" = 1'-0"



LINEN
0.72 M²
7.73 SQ. FT.
BATHROOM #1
7.09 M²
76.33 SQ. FT.
BEDROOM #1
158.3 SQ. FT.
DP1
HT
+4'
Temperature and RH
Aerosol
CO_2 Measurement
CO_2 Injection

Measurements



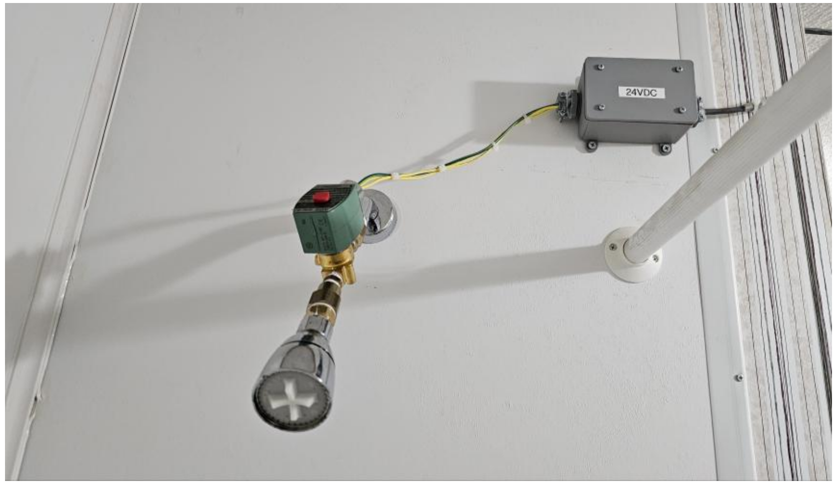
Measurements

- Real-time data collected

Parameter	Number	Instruments
Particle Count	3	QuantAQ, Aerotrak
Carbon dioxide	3	Aranet
Temperature	14	QuantAQ, Vaisala, Aranet, HOBO
Relative Humidity	14	QuantAQ, Vaisala, Aranet, HOBO
Wind Speed	1	Weather Station
Wind Direction	1	Weather Station

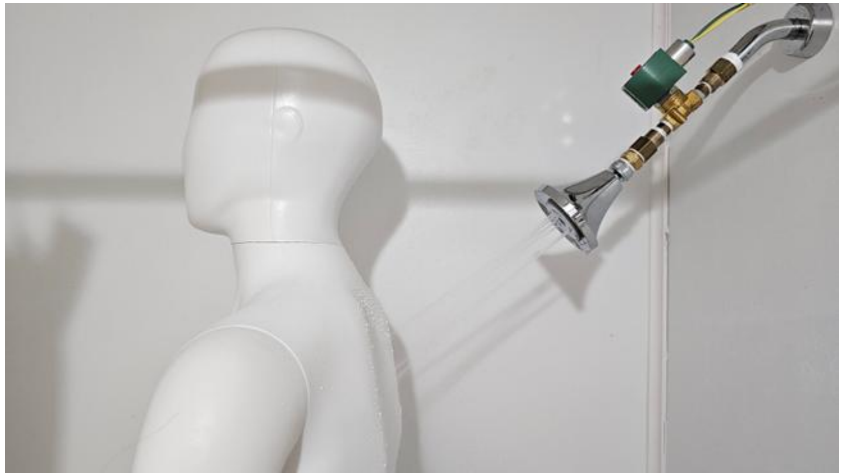
- Spot data
  - Flow rate, water temperature
- Experiments to date
  - > 100

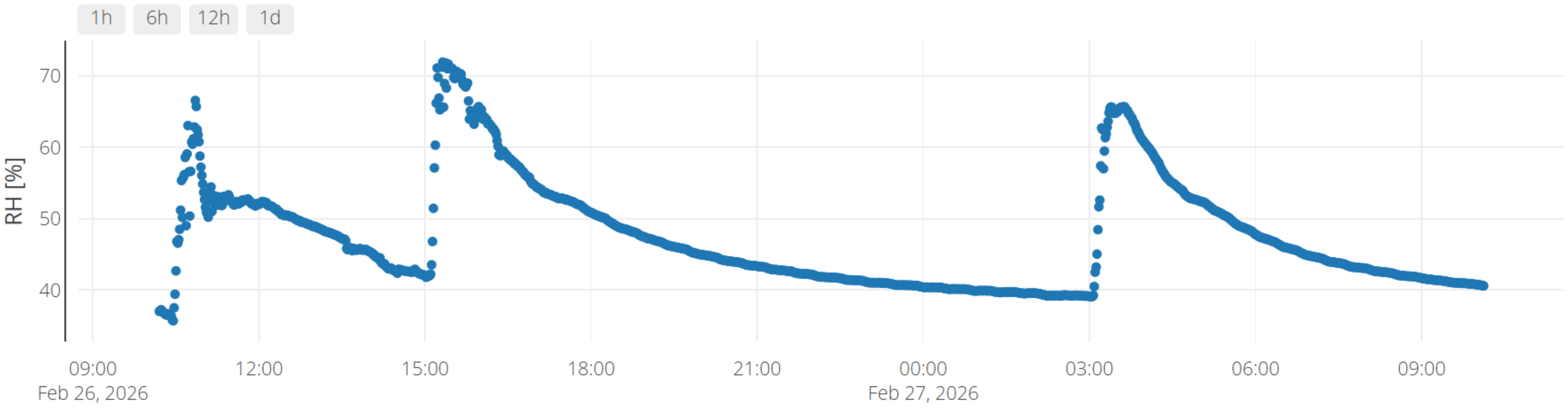
Shower



Shower

- Operated remotely twice a day
  - 3 AM
  - 3 PM
  - *12 h allows RH to return to baseline*

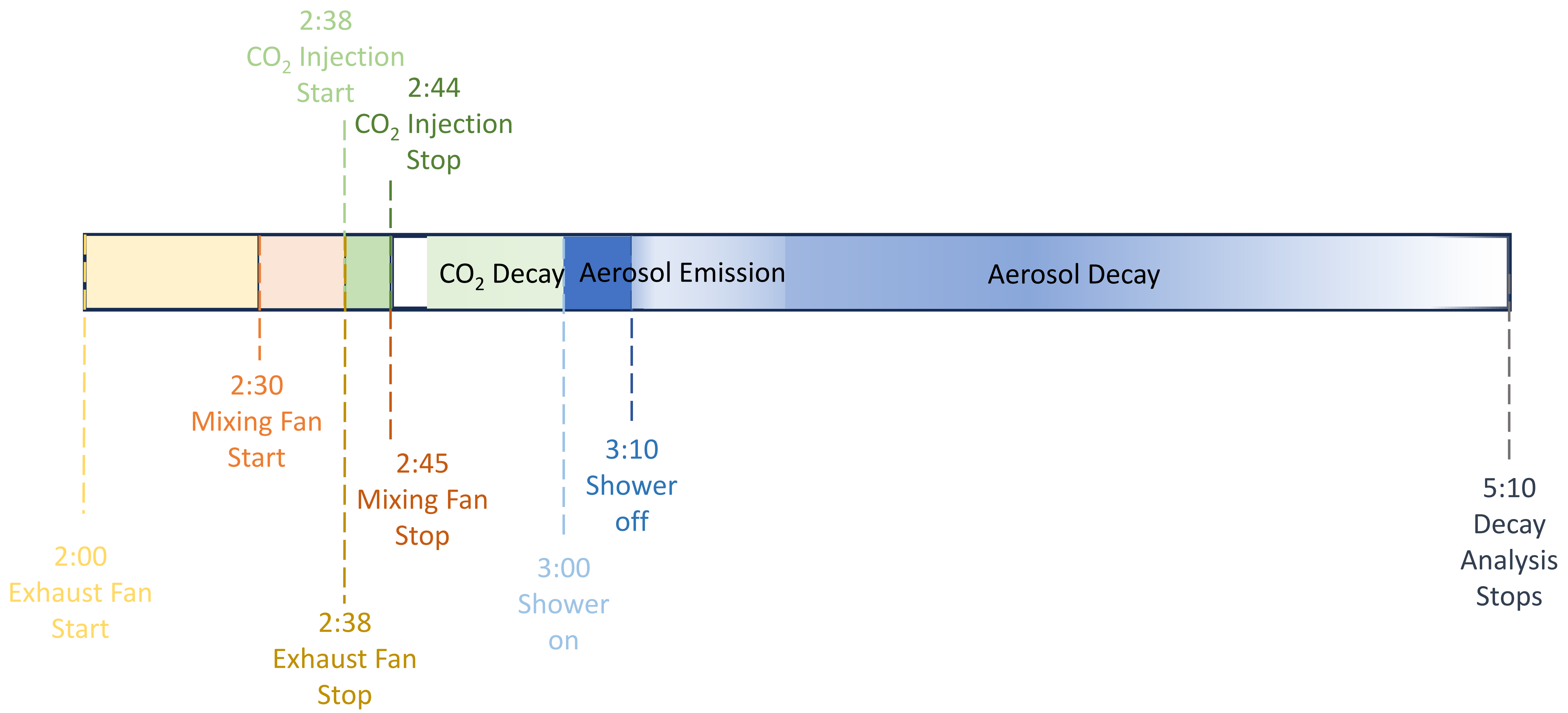


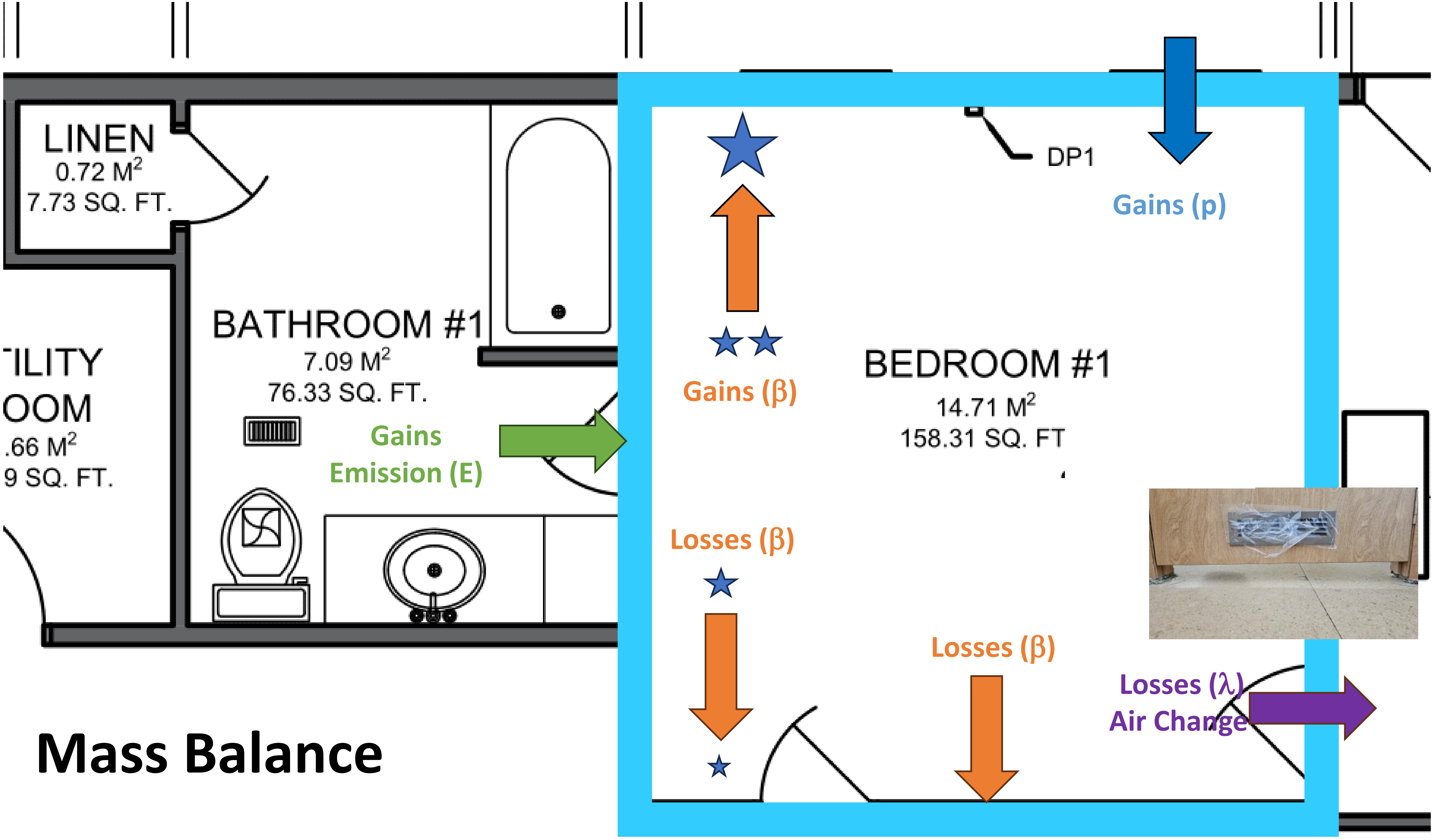


Shower

- Flow measurement
  - Triplicate mass at beginning and end
  - 5-gallon bucket => 1 min
- Water temperature
  - Triplicate at beginning and end

Experimental Timeline



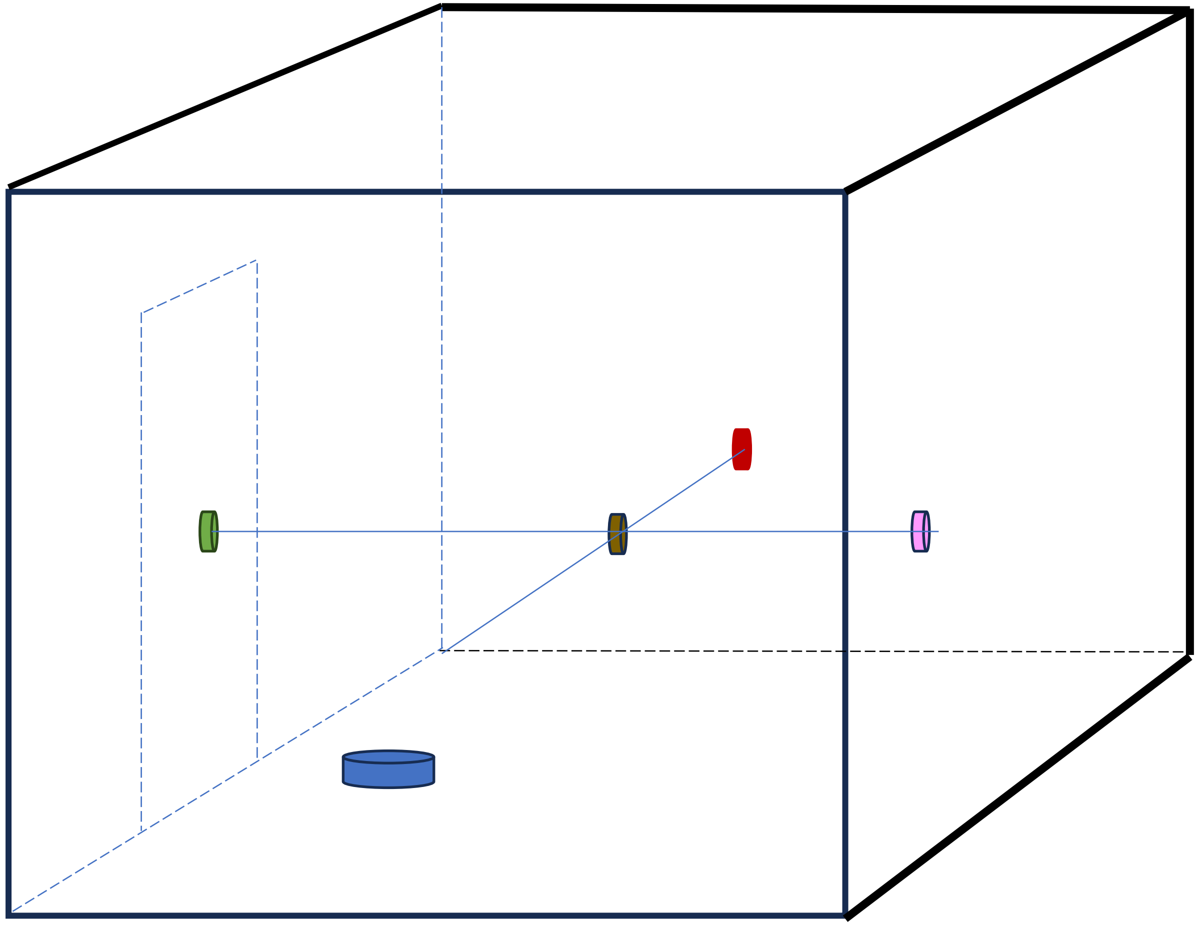


Mass Balance

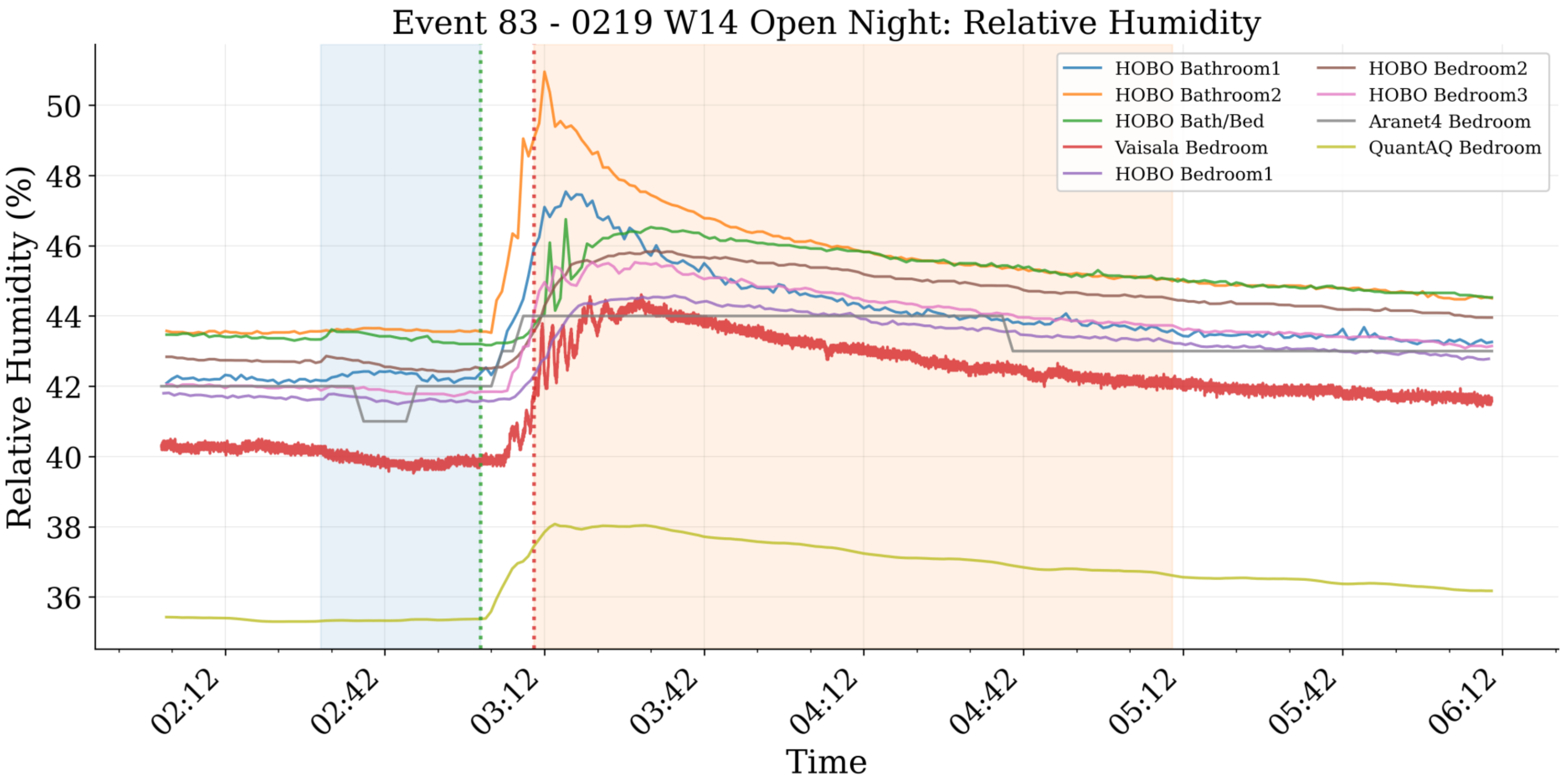
Analysis Assumptions

- **Well mixed?**
- **Air change (λ)**
  - Constant for each experiment and represented by CO_2 decay
- **Other processes rate (β)**
  - Constant in each experiment
- **Emission rate (E)**
  - Constant and represented by particles entering from bathroom

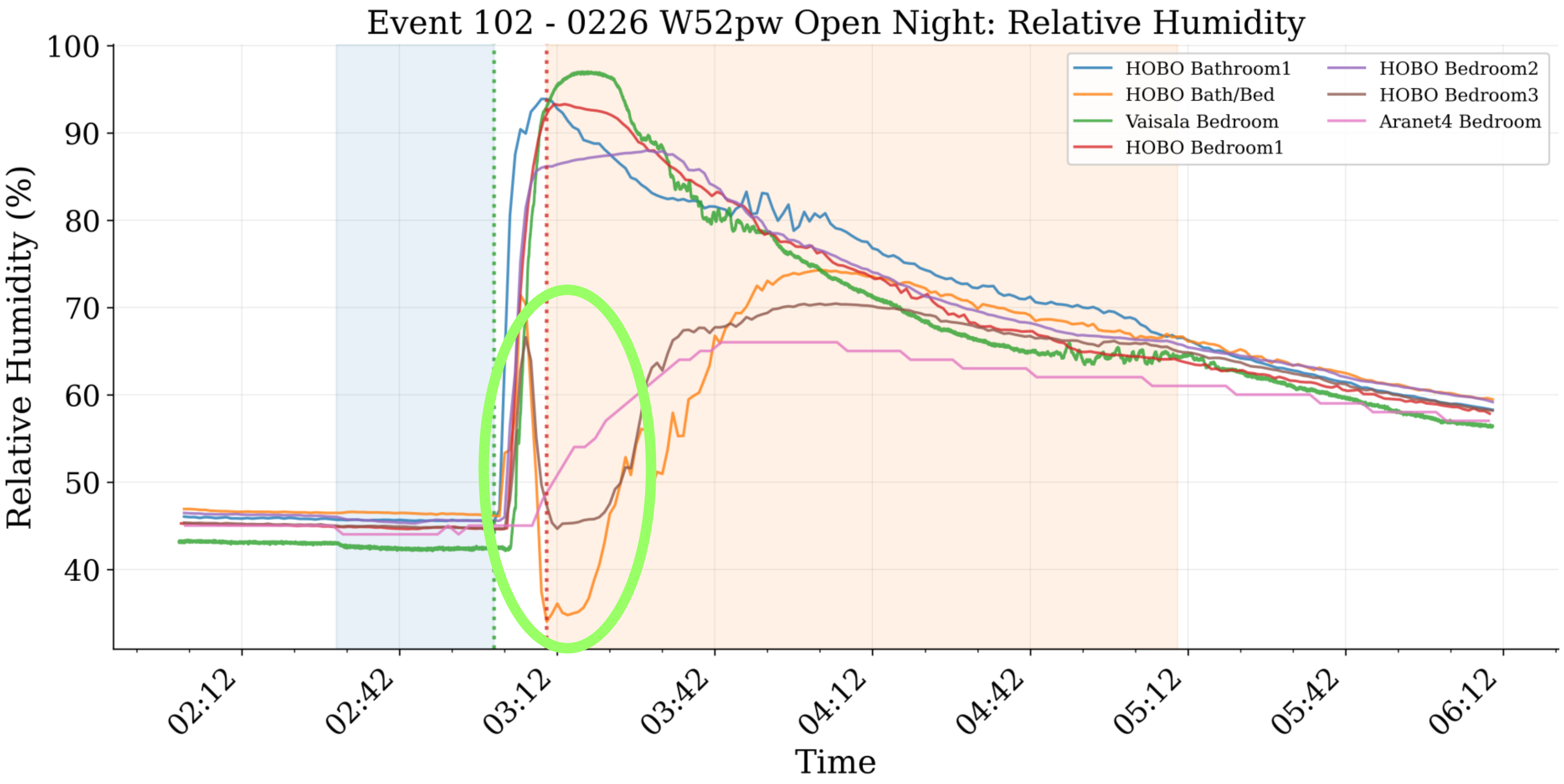
Well Mixed: Preliminary Analysis



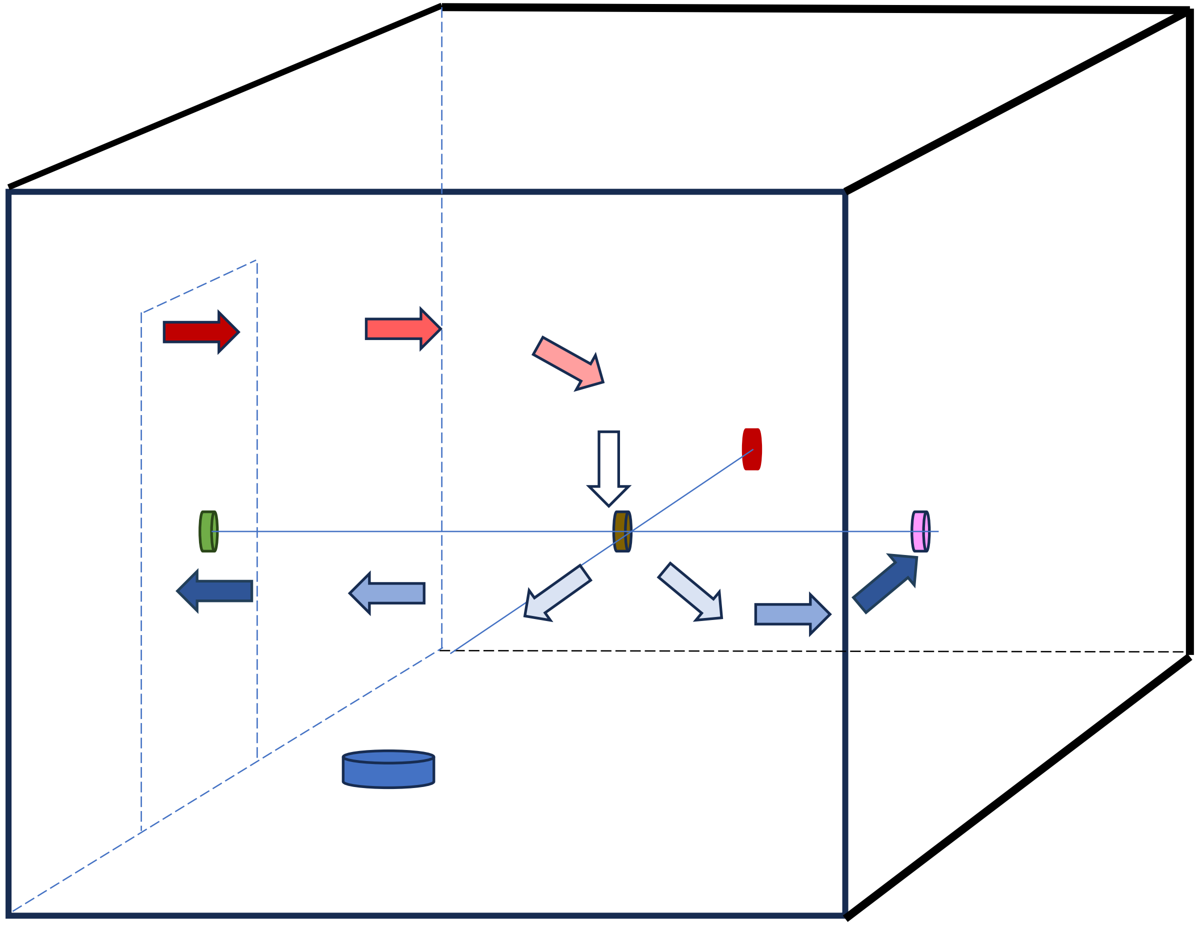
Well Mixed: Preliminary Analysis: 14 C



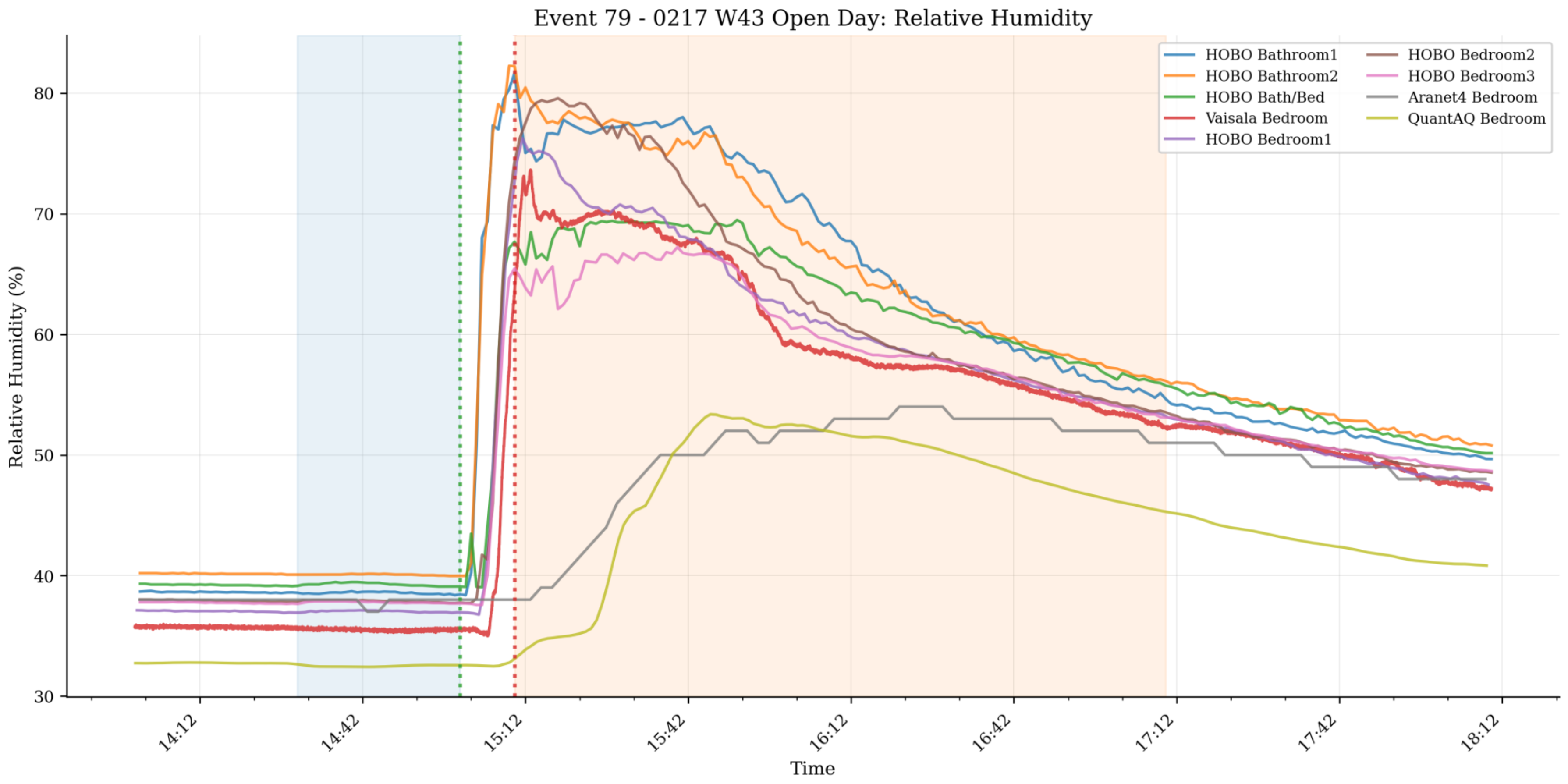
Well Mixed: Preliminary Analysis: 53 C



Well Mixed: Preliminary Analysis



Well Mixed: Preliminary Analysis: 43 C



Air Change: CO_2 Decay Analysis

- Bedroom mixing fan turned on for 15 minutes
  - (30 minutes prior to shower on to 15 minutes prior to shower on)
- CO_2 is injected for 6 minutes (reaching roughly 1400 ppm)
  - (22 minutes prior to 16 minutes prior to shower on)
- Applied 6-minute rolling average to all data
- Average outdoor and rest of house CO_2 concentration assumed constant during fit
- $C_{t=0}$ is defined as CO_2 in room 5 minutes after injection
- Fit ended after 2 h

Air Change: CO_2 Decay Analysis

- Mass balance:

$$V\frac{dC}{dt} = Q_{out}C_{out} + Q_{Entry}\ C_{Entry} - QC$$

From Outside

From Rest of House

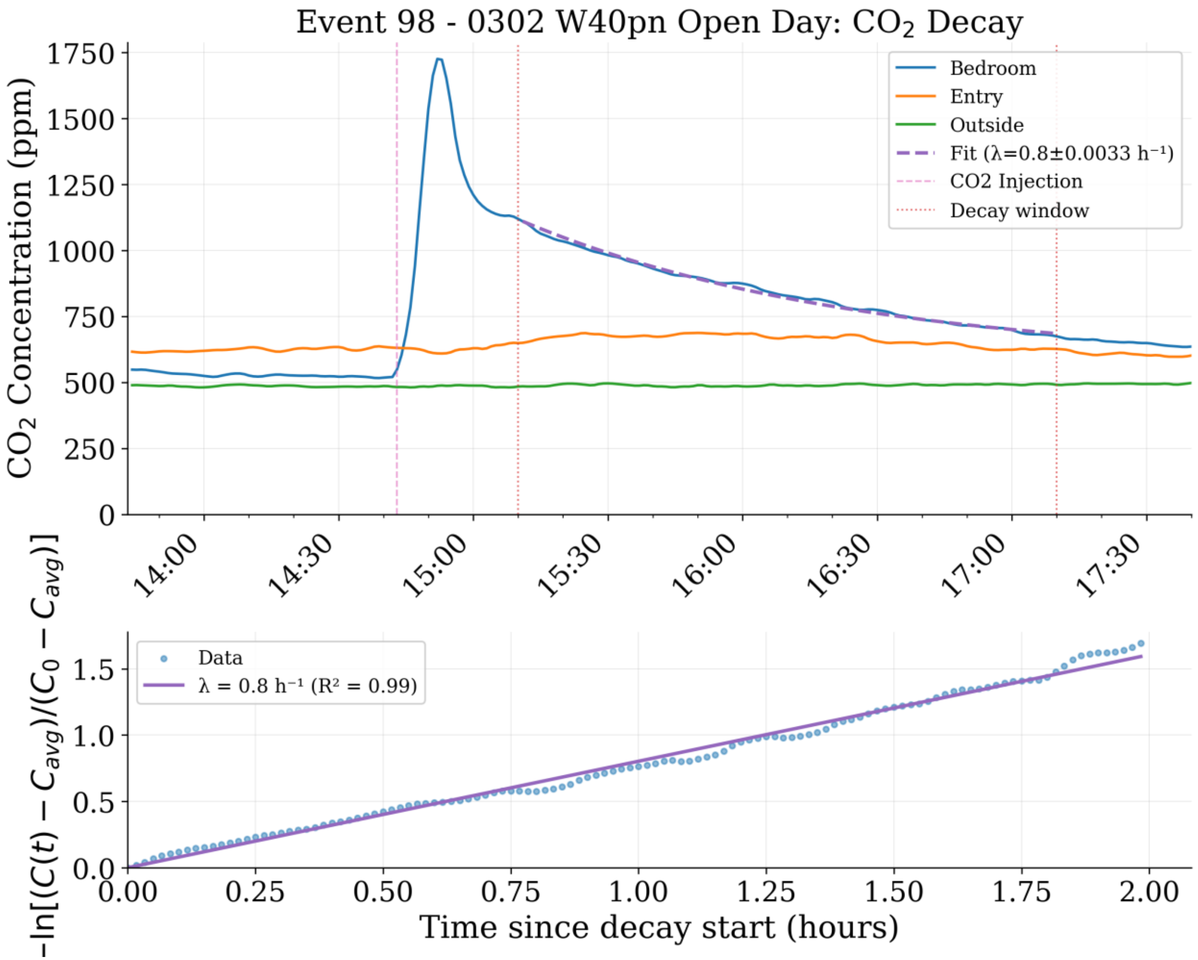
To Rest of House/Outside

- Assume half of flow is from outside/half from rest of house:

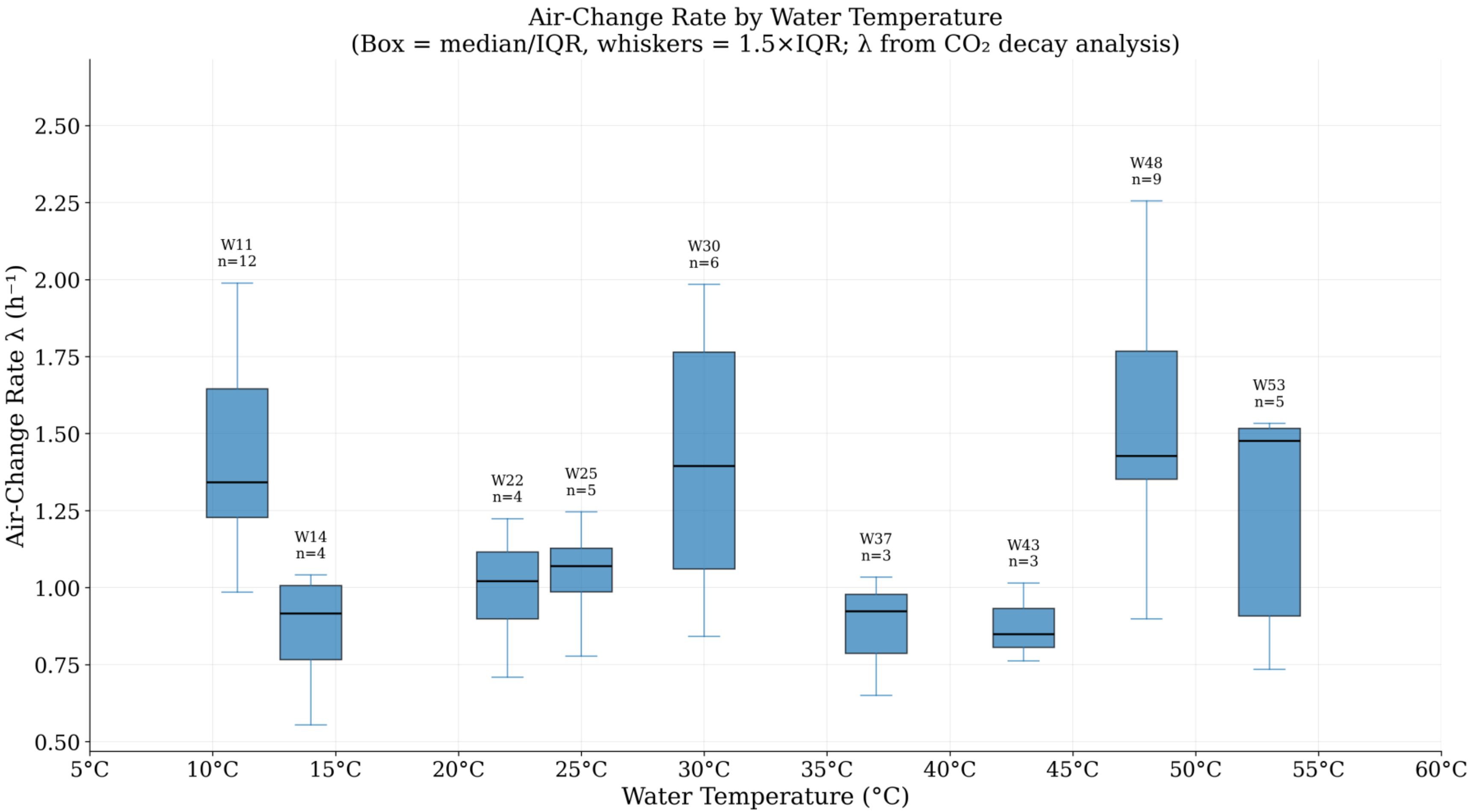
$$V\frac{dC}{dt} = QC_{average,Out\&Entry} - QC$$

$$-\ln\left[\frac{\left(C_{(t)} - C_{average,Out\&Entry}\right)}{\left(C_{t=0} - C_{average,Out\&Entry}\right)}\right] = \lambda t$$

Air Change: CO_2 Decay Analysis



Air Change: CO_2 Decay Analysis



Loss Rate: Aerosol Decay (β) Analysis

- Mass balance after shower:

$$\frac{dC}{dt} = \underbrace{p\lambda C_{out}}_{\text{From house/ outside}} - \underbrace{\lambda C}_{\text{To house/ outside}} - \underbrace{\beta C}_{\text{Aerosol growth/ deposition}}$$

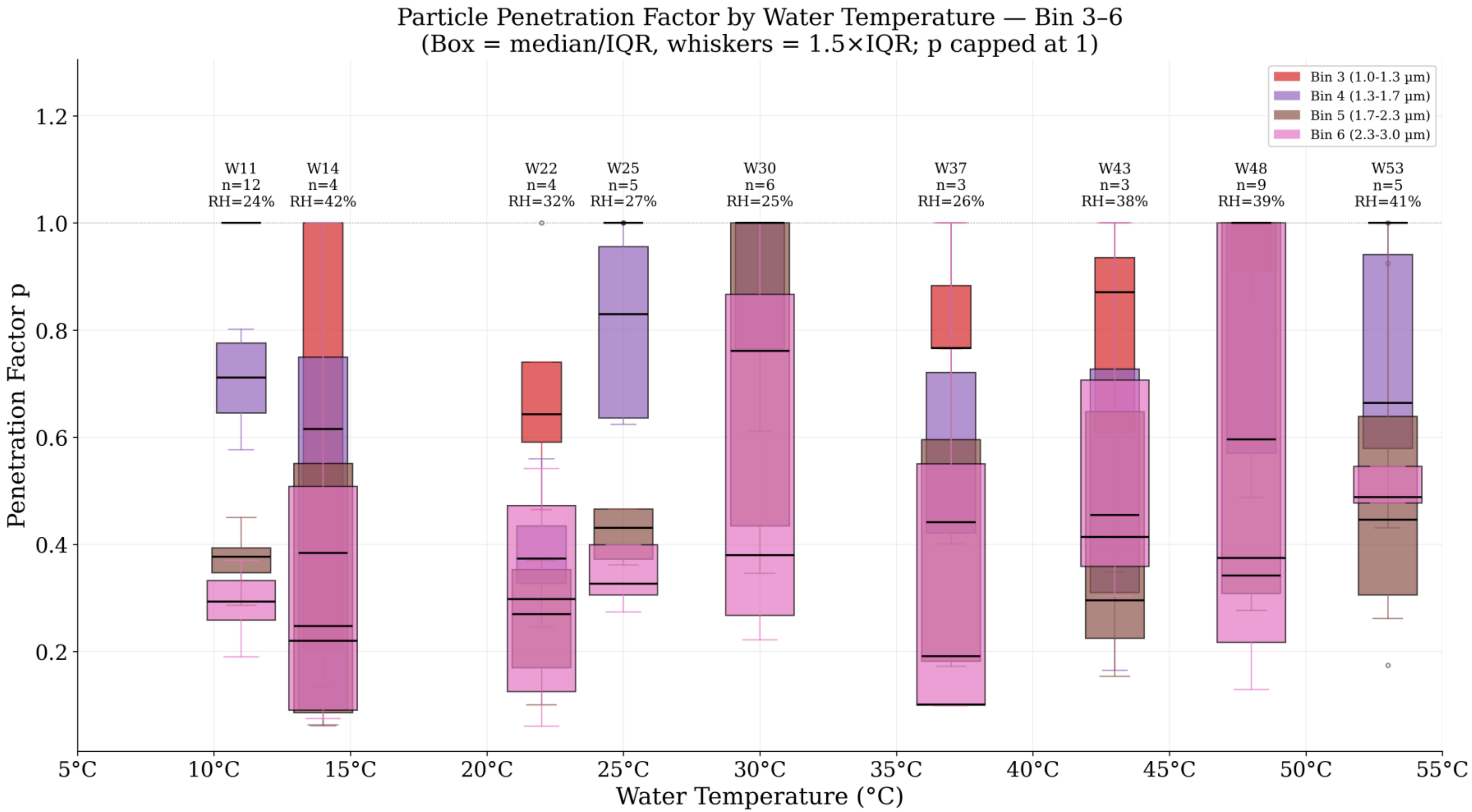
- Air change (λ) from CO_2 injection prior to shower.
- Penetration factor (p)

$$p = \frac{C}{C_{out}}$$

  - Calculated from average data 6 hour prior to 1 h prior to shower
  - Limited to 0 to 1. Often set to 1 when indoor larger than outdoor.

Bin Sizes (Microns)
0.35-0.46
0.46-0.66
0.66-1.0
1.0-1.3
1.3-1.7
1.7-2.3
2.3-3

Loss Rate: Aerosol Decay (P) Analysis



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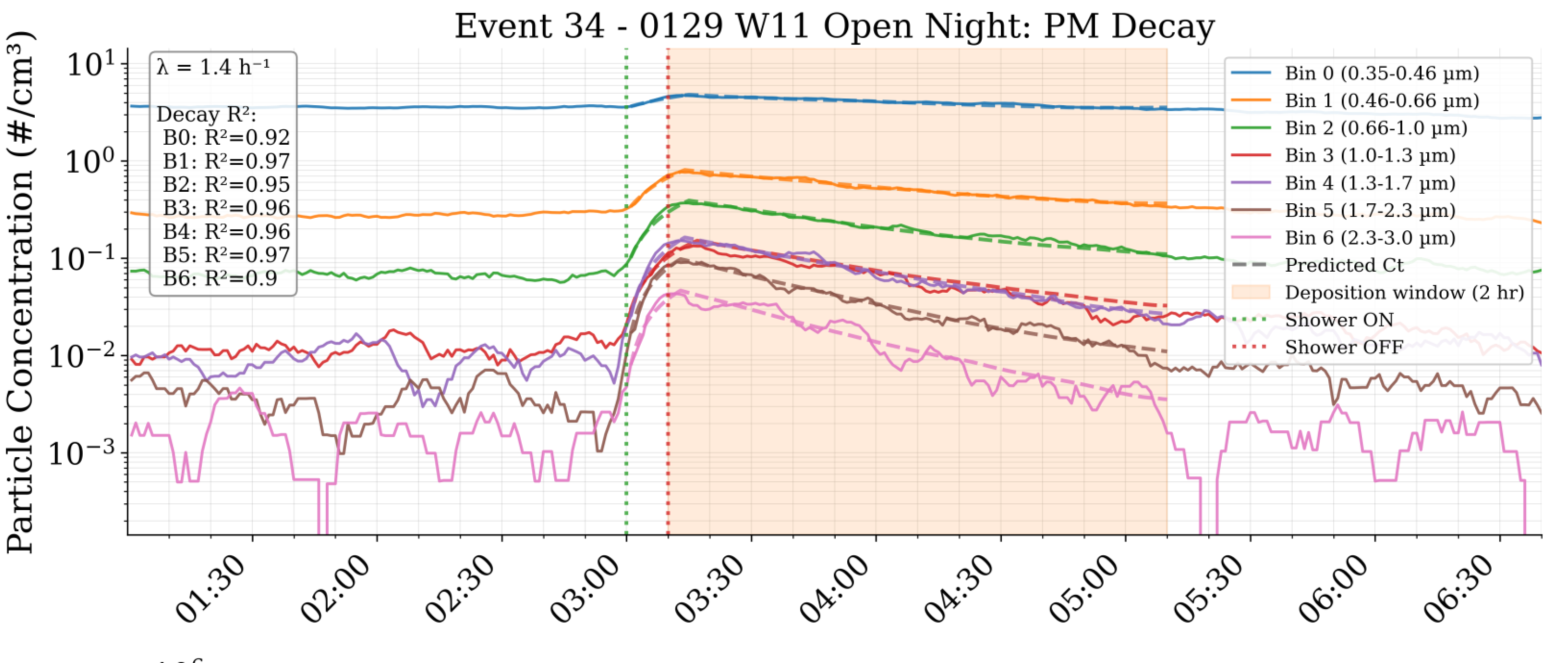
Loss Rate: Aerosol Decay (β) Analysis

- β determined with best fit of data from peak concentration until 2 h after shower was turned on.
- β can be negative (growth in bin size)



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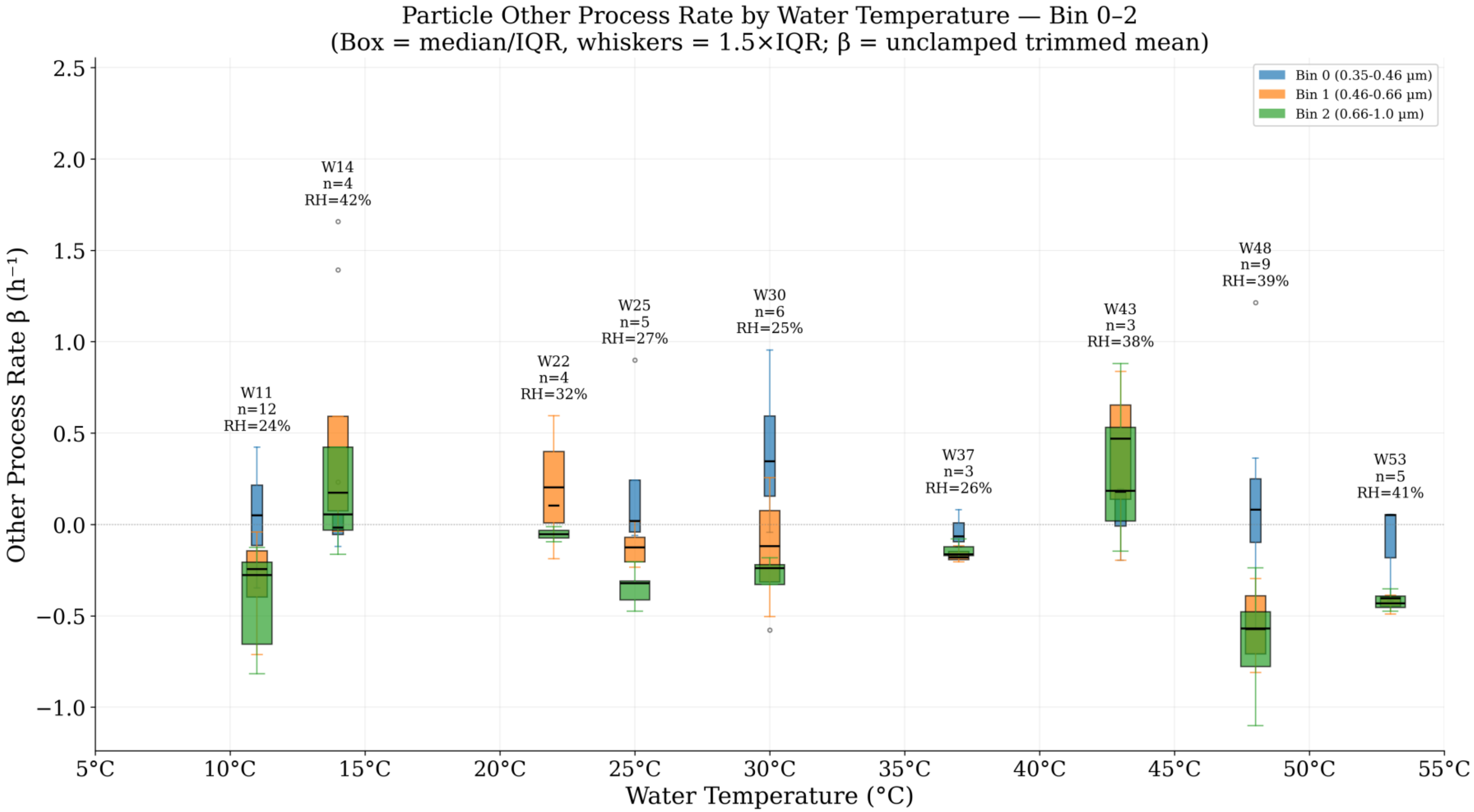
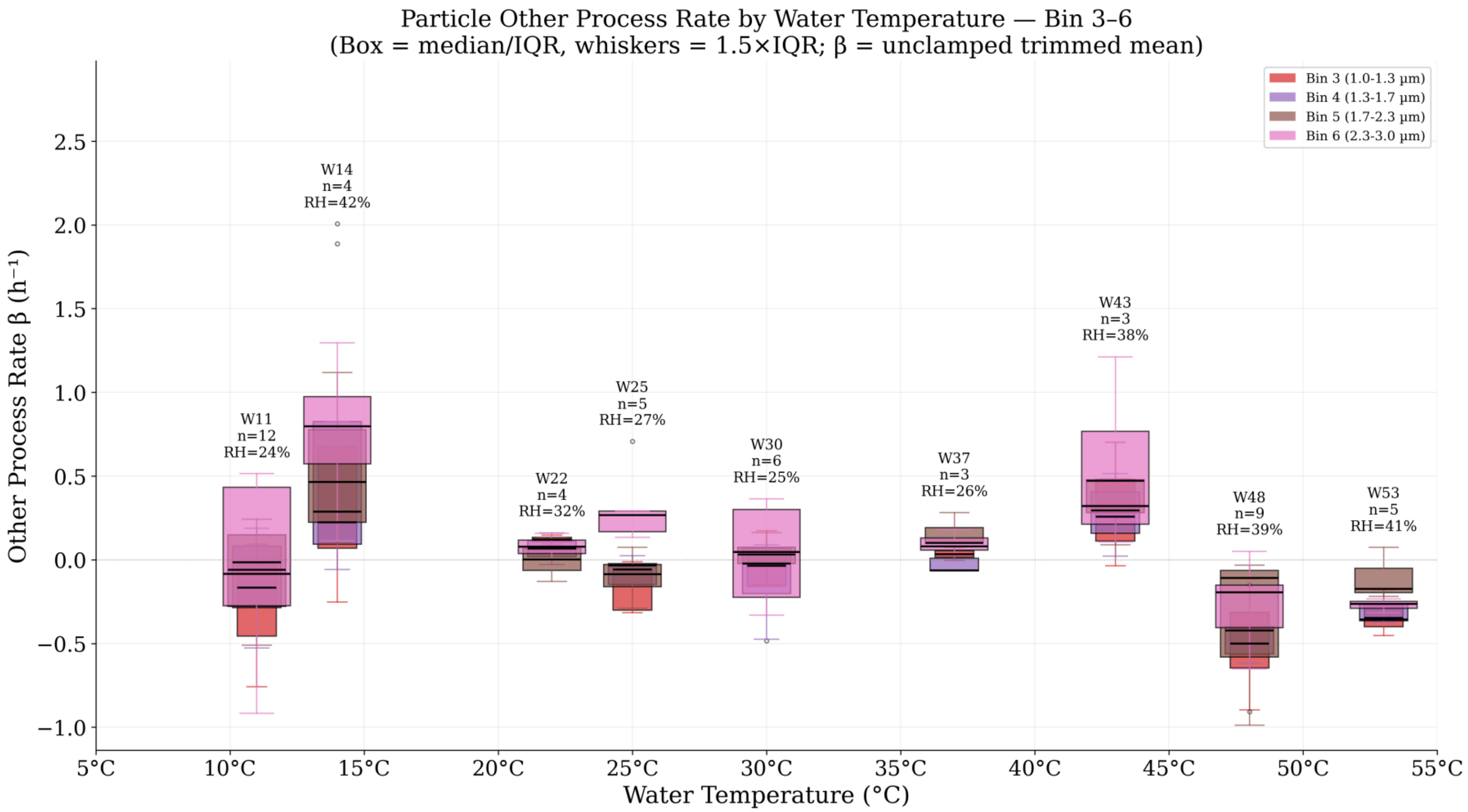


Loss Rate: Aerosol Decay (β) Analysis



Emission Rate: Aerosol Analysis

- Mass balance during shower:

$$\frac{dC}{dt} = \underbrace{p\lambda C_{out}}_{\text{From house/ outside}} - \underbrace{\lambda C}_{\text{To house /outside}} - \underbrace{\beta C}_{\text{Aerosol growth/ deposition}} + \underbrace{\frac{\boxed{E_t}}{V}}_{\text{Aerosol Emitted from Bathroom}}$$

- Air change (λ) from CO_2 injection prior to shower
- Penetration factor (p) from data prior to shower on
- Aerosol loss rate (β) from decay period after peak concentration

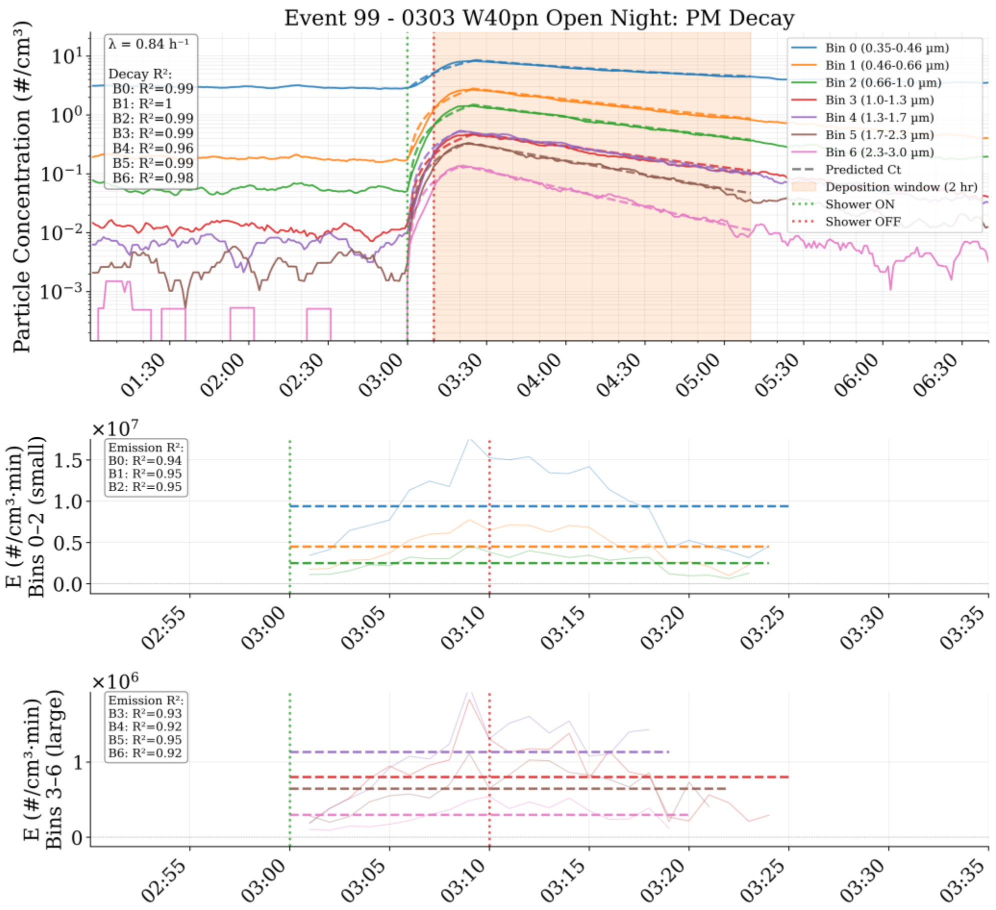
Emission Rate: Aerosol Analysis

- Emission rate calculated numerically:

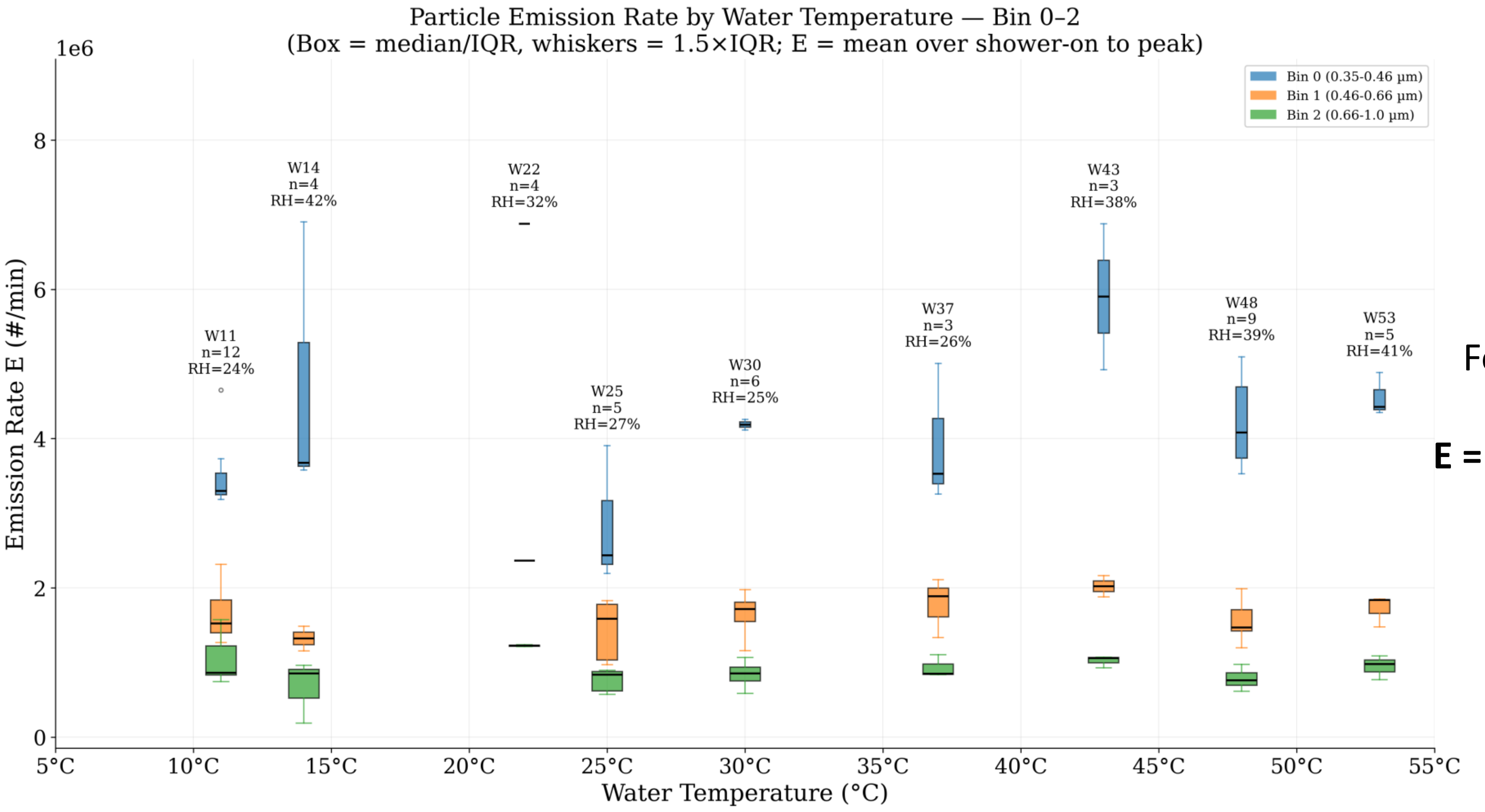
$$E_t = p\lambda V C_{out,t} + \frac{V\left(C_t - C_{t(i+1)}\right)}{\Delta t} - \lambda V C_t - \beta V C_t$$

- Emission rate determined from shower on time until peak bedroom concentration in bin size

Emission Rate: Aerosol Analysis



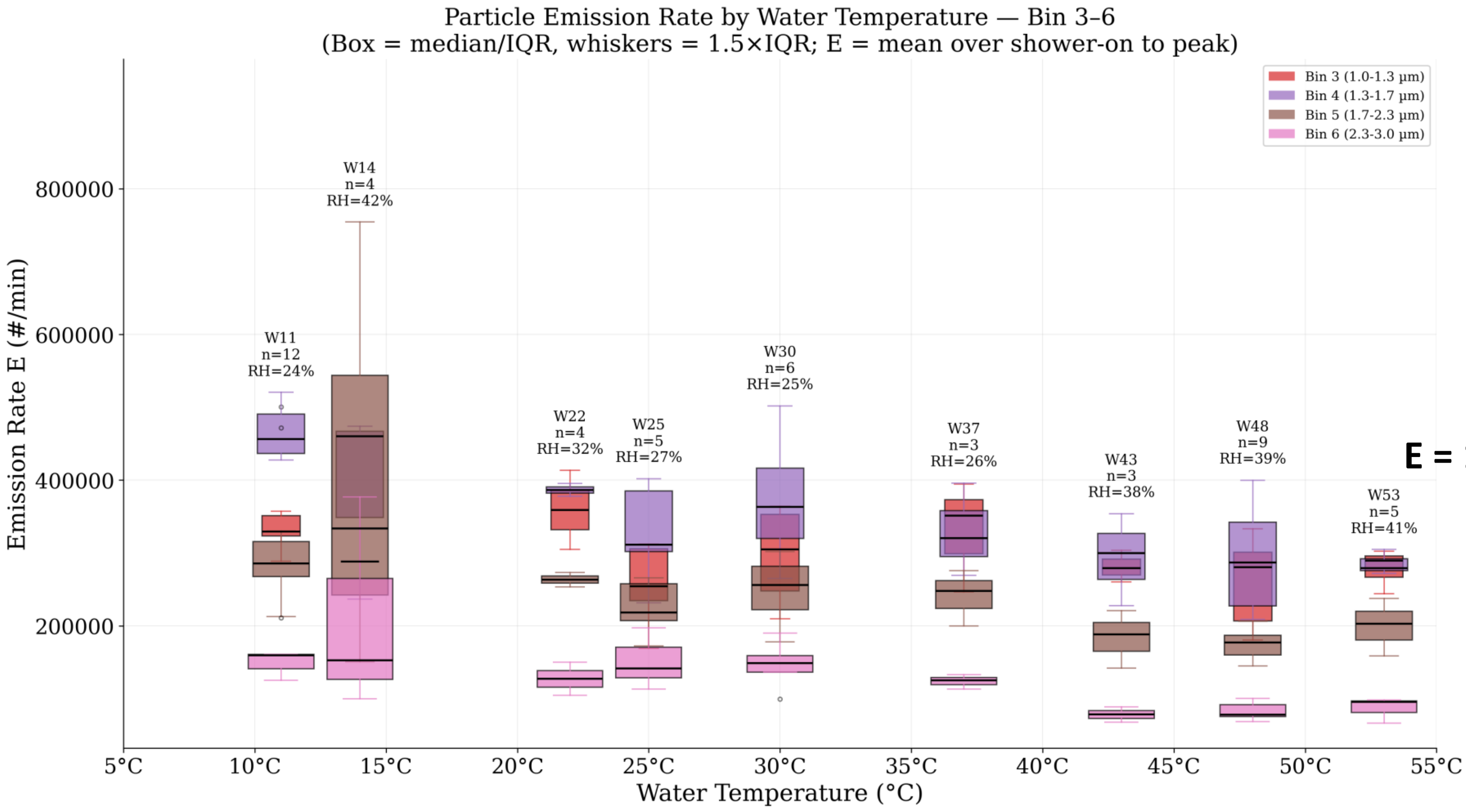
Average Emission Rate: Aerosol Analysis



For particles below 1 micron:

E = 10^5 to 10^6 #/min

Average Emission Rate: Aerosol Analysis



For particles
1-3 microns:
E = 10^5 to 10^6 #/min

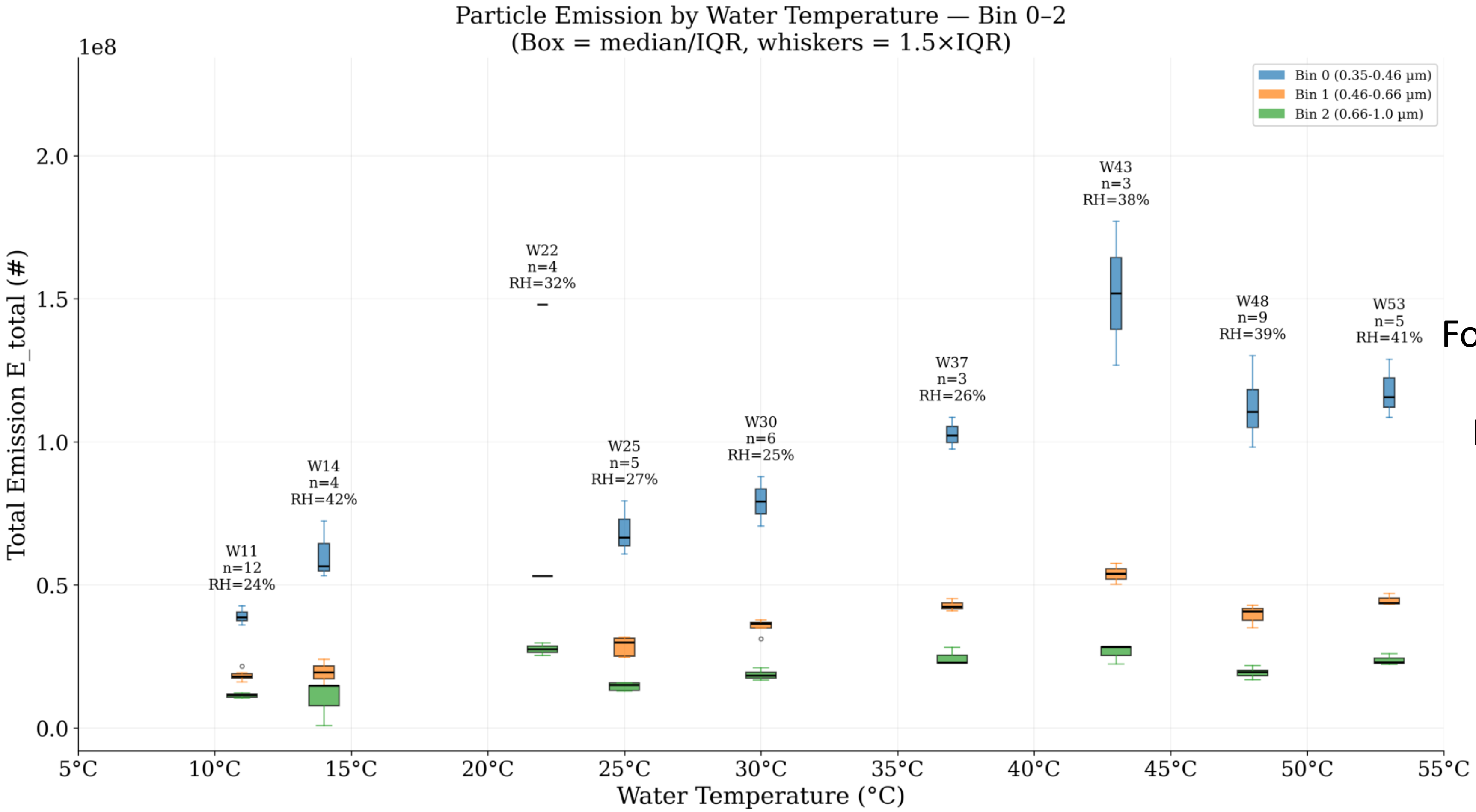
Total Emission

- Sum of emission rates:

$$E_{total} = \Delta t * \sum \frac{E_t + E_{t+1}}{2}$$

- Summed from shower on until peak concentration in bin size.

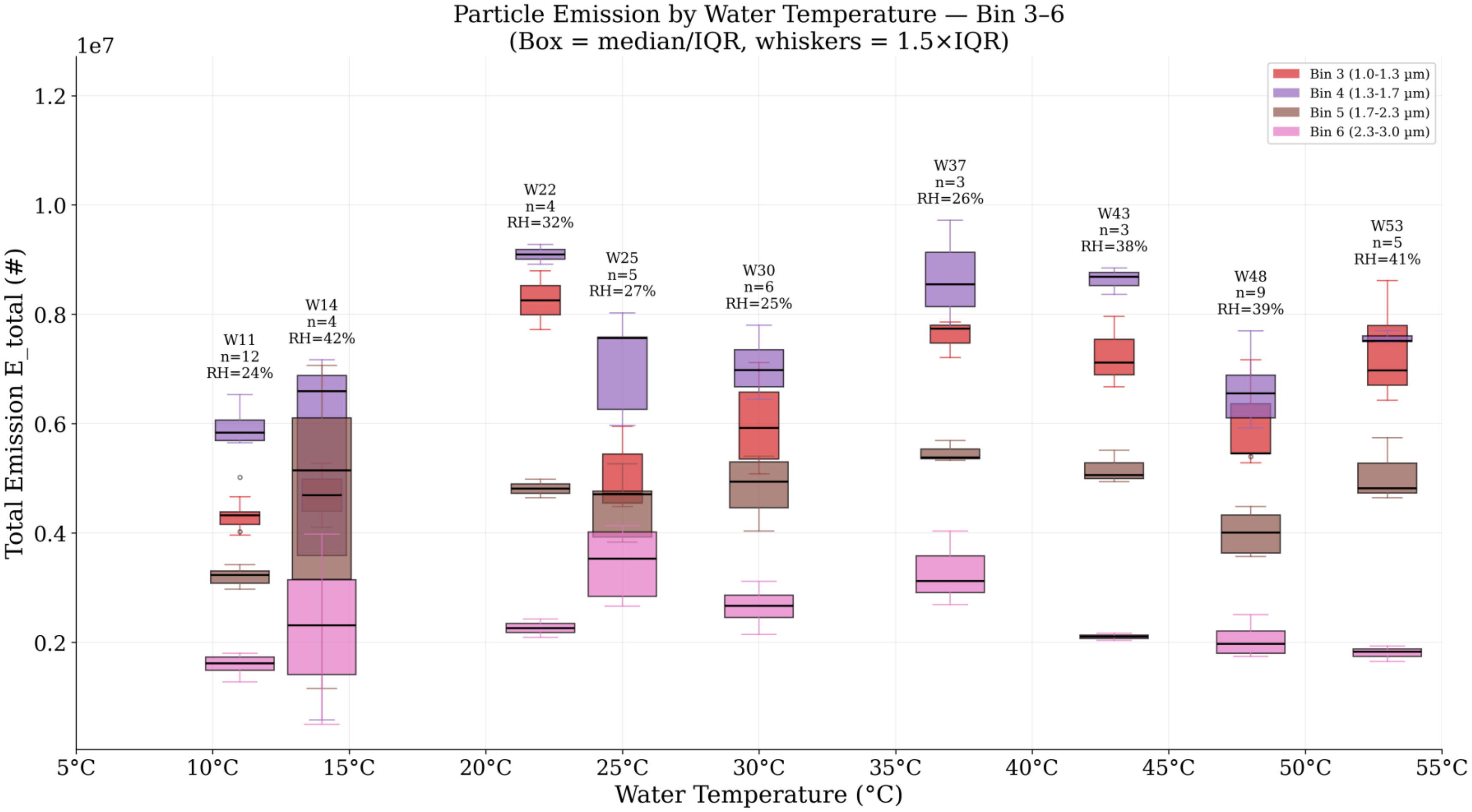
Preliminary Results: Water Temperature



For particles below 1 micron:

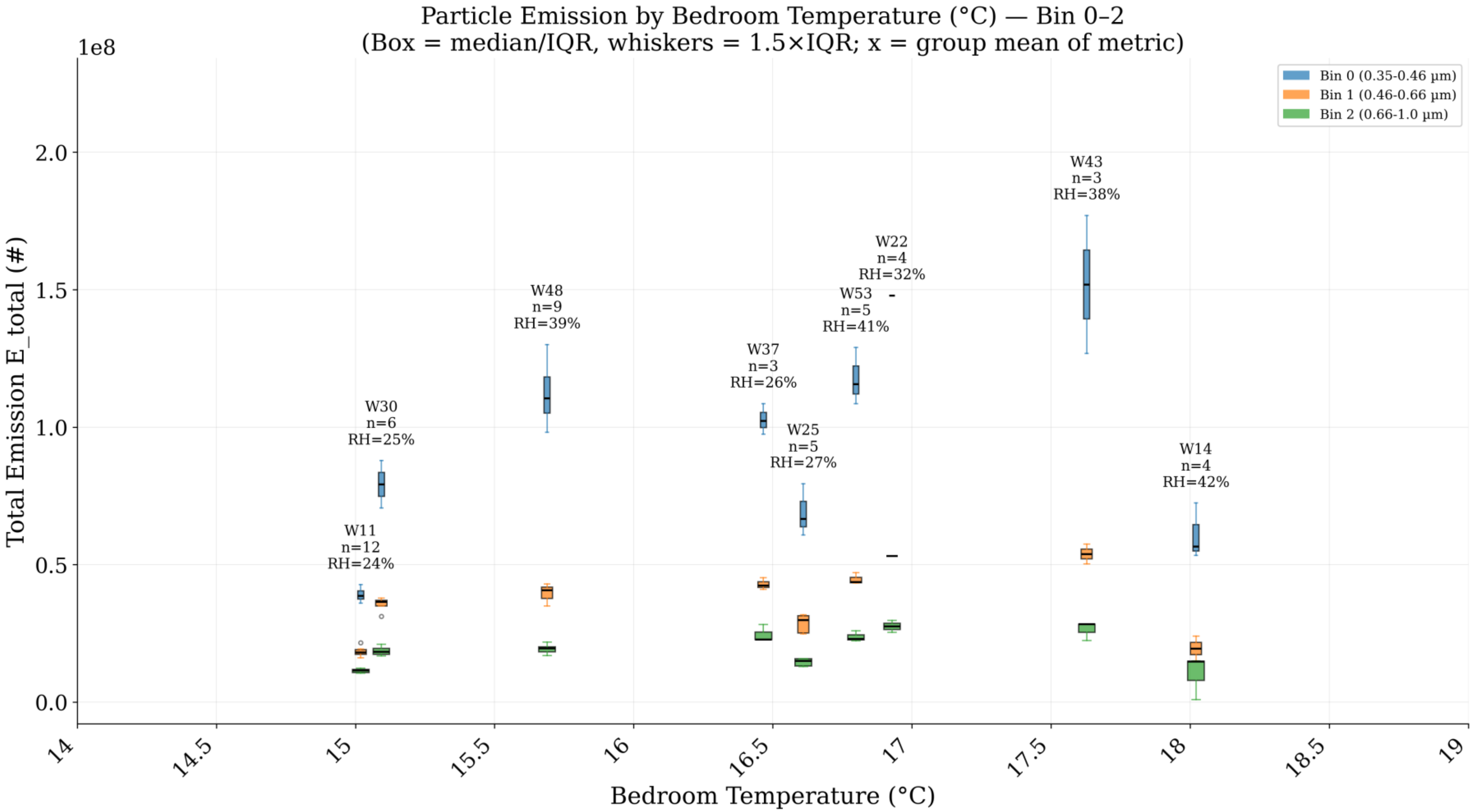
E = 10^7 to 10^8 #

Preliminary Results: Water Temperature

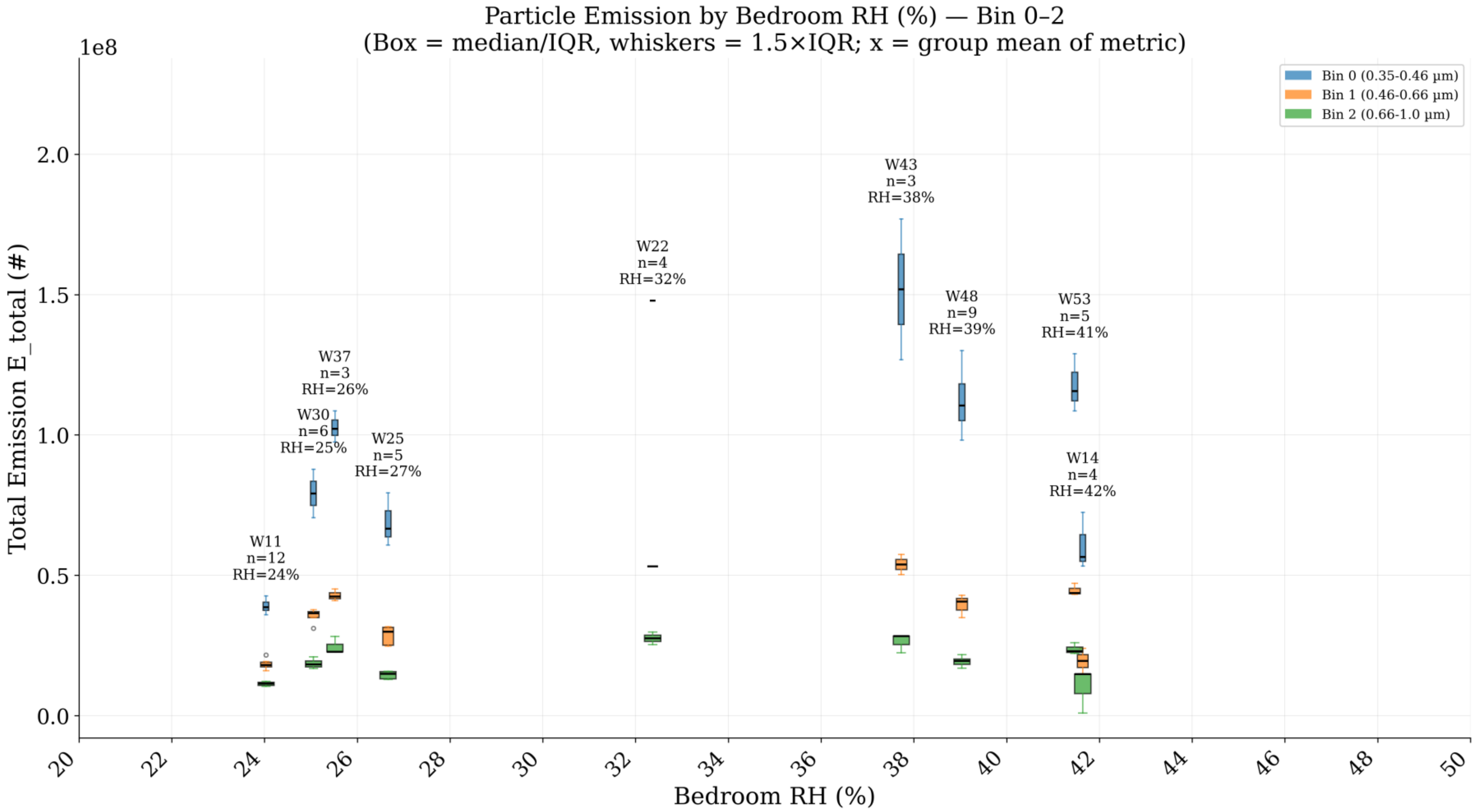


For particles
1-3 microns:
E = 10^6 to 10^7 #

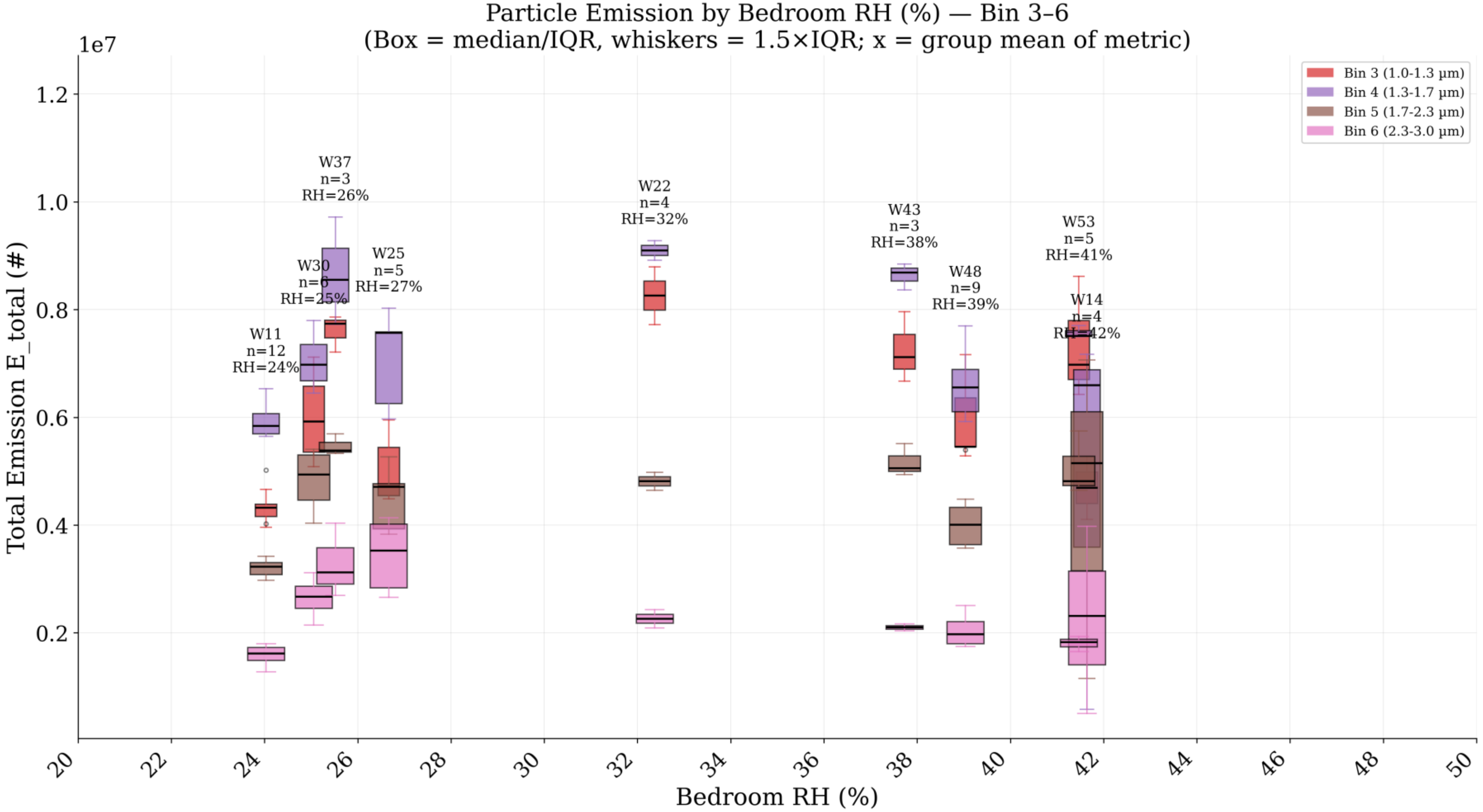
Preliminary Results: Air Temperature



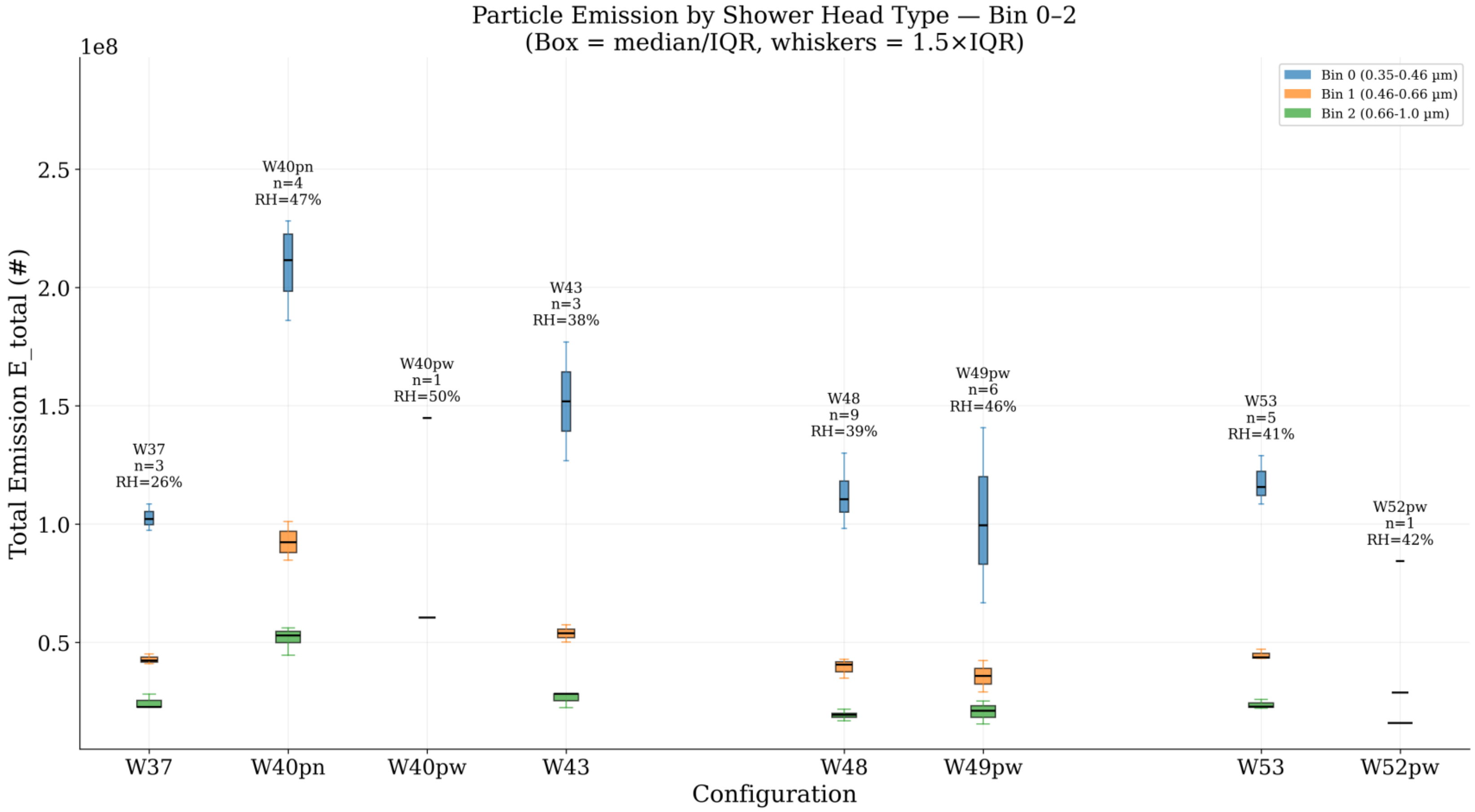
Preliminary Results: Initial RH



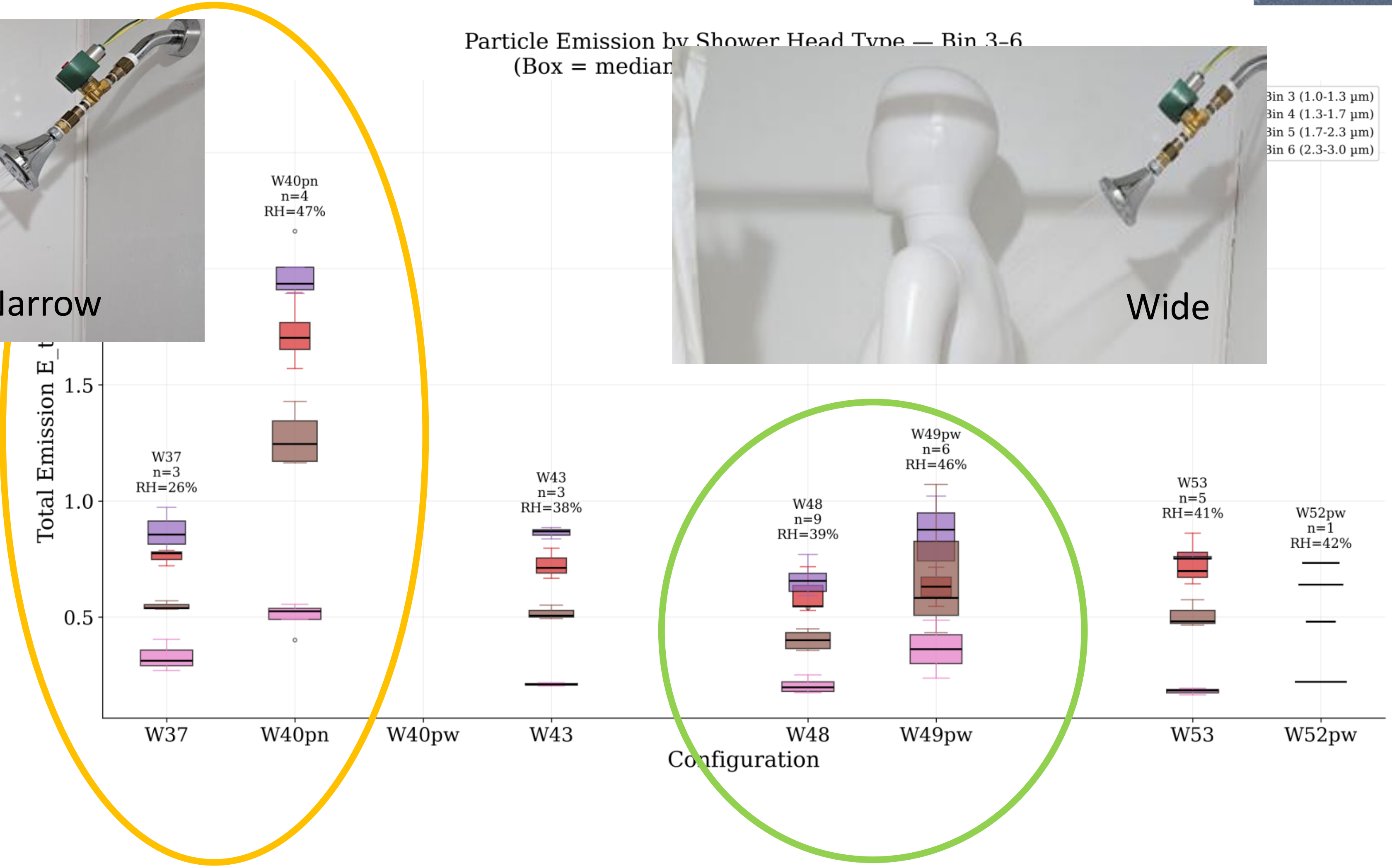
Preliminary Results: Initial RH



Preliminary Results: Shower head



Preliminary Results: Shower head



Next Steps

- Shower head settings? Other shower heads?
- Manny Quin
- Flows
- Ventilation fan
- Location of QuantPM, number of QuantPMs

Questions?